Machine Learning 101 for Financial Trading

FSE-DAT-3019-1.1 | Data Analytics and Computational Modelling

Sector: Financial Services | Proficiency Level 3

FACILITATOR GUIDE

Public edition - assessor keys withheld

Tertiary Infotech Academy Pte Ltd

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Version History

Version No	Date Effected	Change Update	Approved by
2.3	22 October 2026	Assessment papers: every Practical Performance sub-question now has its own answer box, with screenshot boxes naming the lab where the step was practised, and the Written Assessment's ruled lines are replaced by answer boxes, so candidates type or paste their answers into the paper and upload it to the LMS. The PP no longer names the colleague's fault in its evidence list or the program settings, and the Assessment Plan's Annex A prints the K-code on every question.	Dr Alfred Ang
2.2	22 October 2026	Screenshots added to Labs 7, 8 and 9: a Bollinger Band mean-reversion strategy in TradingView with the Paper Trading order panel (Lab 7), the simulator's paper order ticket and its passed pre-trade risk checks and fill (Lab 8), and Performance Analytics comparing a strategy with SPY buy-and-hold (Lab 9). Slide references in the Lesson Plan, Learner Guide and Facilitator Guide updated to the new slide numbers.	Dr Alfred Ang
2.1	21 October 2026	Low-code delivery. Every lab now runs in the Trading Simulator without coding, and the Practical Performance is done by configuring and running a provided trading program, with the same scenario, tasks, ability coverage and marking. A Prompt Library provides one AI prompt per lab, with one-click links, for learners who want code generated for them. Getting Started screenshots added (simulator Pipeline Map and Integrations, Alpaca paper dashboard, TradingView chart). Version History added to every document. Topic 2 slides added on mean reversion (the simulator's rules, mean reversion versus trend following, the main risk and safeguards), on the RSI look-back period (RSI(2) versus RSI(14)) and on long and short positions.	Dr Alfred Ang
2.0	18 October 2026	Major revision, focused entirely on quant and algorithmic stock trading. Courseware rebuilt and re-platformed to Alpaca (API-first broker, paper trading, 24/5 US stock trading) and TradingView (charting, Pine Script v6, Strategy Tester, Paper Trading) with Python. Four topics: quant success stories and the evolution of quant from Thorp and Renaissance through statistical arbitrage to high-frequency trading, the paradigm shift from machine learning to LLMs and its impact on quant, and AI agents versus agentic AI in trading; the classification of trading algorithms, indicators, strategies, a TradingView versus Alpaca comparison, order placement and metric definitions; machine learning models, statistical modelling, A/B testing and pairs trading as a hedging algorithm; and an end-to-end, automated algorithmic trading pipeline. 'How the algorithm works' slides explain each algorithm on real market data. Twenty hands-on labs build one pipeline stage by stage, each with an instruction PDF containing step-by-step AI prompts, and include live paper trading of a strategy. Assessment: Written Assessment (short-answer questions, K1-K9) and Practical Performance (four tasks, A1-A10) with a standalone SMART marking rubric. Facilitator Guide added. Fee restated in SGD exclusive of GST, optimum class size stated, provider name updated to Tertiary Infotech Academy Pte Ltd.	Dr Alfred Ang

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Version No	Date Effected	Change Update	Approved by
1.1	22 December 2022	Multiple choice questions removed; practical test retained as the sole assessment instrument.	Dr Alfred Ang
1.0	10 June 2022	Initial release for pilot launch, aligned to Data Analytics and Computational Modelling (Proficiency Level 3).	Dr Alfred Ang

This document is at Version 2.3, effective 22 October 2026.

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1. About This Guide

This public edition of the Facilitator Guide omits the assessor's keys. This Facilitator Guide supports the trainer and assessor in delivering Machine Learning 101 for Financial Trading. It sets out the course information, the daily lesson plan with slide references, facilitation notes and timing cues for every topic and lab, the administration of the assessment and its timings. The assessor's keys, model answers and rubric are held in the trainer-only full edition and the Assessment Plan.

Item	Details
Course Title	Machine Learning 101 for Financial Trading
TSC Code and Title	FSE-DAT-3019-1.1 Data Analytics and Computational Modelling
Proficiency Level	3 - Identify and utilise appropriate statistical algorithms and data models to perform data analyses, test hypotheses and derive inferences, patterns or solutions from data
Course Duration	16 hours over 2 training days (14 hours classroom training + 2 hours assessment)
Daily Schedule	0900 - 1830 (8 instructional hours per day, excluding two 15-minute tea breaks and a 1-hour lunch break)
Mode of Delivery	Instructor-led classroom training with hands-on practical laboratory exercises on the Alpaca paper-trading platform, TradingView and the low-code Trading Simulator
Class Size / Ratio	Optimum class size of 20 participants (maximum 20) | Trainer : learner 1 : 20
Assessment	Written Assessment (45 min, K1-K9) | Practical Performance (75 min, A1-A10)
Funding	IBF Standards Training Scheme (IBF-STS), subject to eligibility; at least 75% attendance and passing the assessment are required
Companion documents	Lesson Plan, Trainer Slides, Learner Guide, Assessment Plan, Assessment Rubric, lab folder (labs)

Learning outcomes

LO	Learning Outcome	TSC Ref
LO1	Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.	A1, A6, K1, K9
LO2	Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.	A2, A3, K3, K5, K7
LO3	Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.	A4, A5, K2, K4, K6
LO4	Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.	A7, A8, A9, A10, K7, K8

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2. Preparing to Deliver

Two weeks before the class

- Send the joining instructions: bring a laptop with the Trading Simulator installed (Learner Guide setup guide), create a free Alpaca account and a free TradingView account in advance if possible.
- Confirm the venue network allows outbound HTTPS to paper-api.alpaca.markets, data.alpaca.markets and tradingview.com. If it does not, plan to run every lab in offline mode - the bundled data and simulator make this seamless.
- Install the Trading Simulator on the trainer laptop and run every lab once on its Course Labs page (install steps: Learner Guide and labs_lowcode/Lab01-alpaca-setup/Lab01_Instructions.pdf). Each lab folder holds its LabNN_Instructions.pdf with the simulator steps and checks.

On the day

- Check that every learner's Trading Simulator opens at http://localhost:8000 before Lab 1.
- Log in to the trainer Alpaca paper account and TradingView for live demonstrations; never use a live-money account.
- Have the Learner Guide open at the setup section; most first-hour issues are simulator installation issues, not course content.
- Prepare the attendance sheet for each session and the assessment attendance record.

Safety and conduct messages to repeat

- Paper trading only; no real money at any point.
- Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.
- API keys are credentials: never put them on a slide, in a chat, or in a file that is shared.

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3. Daily Lesson Plan

Identical to the Lesson Plan; the Slides column is read from the built deck.

Day 1

Time	Duration	Topic / Activity	LO	TSC Ref	Slides
09:00 - 09:30	30 min	Course Overview, Trainer Introduction and Ground Rules	-	-	1-34
09:30 - 10:45	75 min	T1.1-1.4 Quant Success Stories, the Evolution of Quant, the ML-to-LLM Shift, and Agentic AI versus AI Agents	LO1	K1, K9	36-58
10:45 - 11:00	Tea Break				
11:00 - 11:30	30 min	T1.5-1.6 Forms of Financial Data, Big Data Applications and Trading Hypotheses	LO1	A1, A6, K1, K9	59-66
11:30 - 11:50	20 min	Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account	LO1	K5	67
11:50 - 12:10	20 min	Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView	LO1	A3, K5, K9	69
12:10 - 12:30	20 min	Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality	LO1	A6, K9	71
12:30 - 13:30	Lunch Break				
13:30 - 13:45	15 min	Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test	LO1	A1, A6, K1	74
13:45 - 14:30	45 min	T2.1-2.2 Technical Indicators and Standard Algorithmic Trading Strategies	LO2	A2, K3	78-90, 96-118
14:30 - 14:55	25 min	Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins	LO2	A3, K3, K7	124
14:55 - 15:25	30 min	Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation	LO2	A2, A3, K3, K7	127
15:25 - 15:40	Tea Break				
15:40 - 16:05	25 min	Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals	LO2	A2, K3, K7	131
16:05 - 16:50	45 min	T2.3-2.4 Analytics Platforms, Coding Languages and Order Placement	LO2	A3, K5, K7	91-92, 119-123
16:50 - 17:20	30 min	Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session	LO2	A3, K5	134
17:20 - 17:50	30 min	T2.5 Performance and Risk Metric Definitions	LO2	A3, K3	93-95
17:50 - 18:15	25 min	Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table	LO2	A3, K3	138
18:15 - 18:30	15 min	Day 1 Recap and Q&A	-	-	76, 141

Instructional time Day 1: 480 minutes.

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Day 2

Time	Duration	Topic / Activity	LO	TSC Ref	Slides
09:00 - 09:15	15 min	Day 1 Recap and Day 2 Overview	-	-	76, 141
09:15 - 09:40	25 min	T3.1-3.2 Statistical Analyses, Algorithms and Supervised Learning Models	LO3	A4, K2, K4	144-146, 150-152, 157
09:40 - 10:00	20 min	Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay	LO3	A2, A4, K2	161
10:00 - 10:20	20 min	Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position	LO3	A4, K4, K6	163
10:20 - 10:35	Tea Break				
10:35 - 10:55	20 min	T3.3-3.4 Unsupervised Learning and Statistical Modelling Techniques	LO3	A4, K4, K6	147-148, 153-155
10:55 - 11:10	15 min	Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals	LO3	A4, K4, K6	165
11:10 - 11:25	15 min	T3.5-3.6 A/B Testing, and Pairs Trading as a Hedging Algorithm	LO3	A4, A5, K2, K6	149, 156, 158-160
11:25 - 11:40	15 min	Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves	LO3	A5, K2	167
11:40 - 12:00	20 min	Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve	LO3	A4, K6	169
12:00 - 12:30	30 min	T4.1-4.2 Pipeline Architecture and Backtesting with Trading Costs	LO4	A7, A8, K7	173-176, 179
12:30 - 13:30	Lunch Break				
13:30 - 13:55	25 min	Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even	LO4	A7, A8, K7	184
13:55 - 14:05	10 min	T4.3 Forward Testing with Automated Paper Trading	LO4	A7, A8	180-181
14:05 - 14:30	25 min	Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading	LO4	A7, A8, K7	186
14:30 - 14:50	20 min	Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution	LO4	A7, K5, K7	188
14:50 - 15:00	10 min	T4.4 Potential Reasons for Unintended Outcomes	LO4	A9, K8	177, 182
15:00 - 15:20	20 min	Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live	LO4	A9, K8	190
15:20 - 15:35	Tea Break				
15:35 - 15:45	10 min	T4.5 Performance Analysis, Insights and Reporting	LO4	A10	178
15:45 - 16:10	25 min	Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report	LO4	A8, A10	192
16:10 - 16:30	20 min	Lab 20: Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval	LO4	A7, A10, K1	194
16:30 - 17:15	45 min	Written Assessment (Short Answer Questions)	All LOs	K1 - K9	23
17:15 - 18:30	75 min	Practical Performance Assessment	All LOs	A1 - A10	23

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Instructional time Day 2: 480 minutes.

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4. Facilitation Notes by Topic

Topic 1: Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents

Day 1 | Slides 35-76 | A1, A6, K1, K9 | Labs 1-4

Learning outcome: Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.

Key teaching points

- 1.1 Quant Success Stories: What Systematic Trading Has Achieved (K1): Ed Thorp - from blackjack to markets: his fund (Convertible Hedge Associates, 1969; Princeton Newport Partners from 1974) is often called the first market-neutral hedge fund and earned about 20% a year after fees for two decades by hedging warrants and convertibles with probability models.
- 1.2 The Evolution of Quant: From Thorp and Renaissance to High-Frequency Trading (K1, K9): Algorithmic trading is the use of a computer program that follows a defined set of instructions to generate and execute orders automatically, based on price, timing, volume or a mathematical model.
- 1.3 The Paradigm Shift from Machine Learning to LLMs, and Its Impact on Quant (K1, K9): Classical machine-learning quant works on structured data - prices, volumes, fundamentals - with features engineered by the quant (lagged returns, RSI, factor scores) and models such as regression, random forests and gradient boosting fitted to labelled history.
- 1.4 Agentic AI versus AI Agents in Trading, and Multi-Agent Trading Systems (K1): Agentic AI is the capability of AI to plan and act: it calls tools (market-data and broker APIs), uses skills and memory, and follows instructions and guardrails to complete multi-step work when it is asked - for example, an LLM that fetches a stock's bars and news, checks its memory of past trades and proposes an order within its instructions.
- 1.5 Forms of Data within Financial Services and Big Data Applications in Trading (K9, A6): Market data: OHLCV bars (open, high, low, close, volume) at timeframes from one minute to one month; quotes (best bid and ask); trades (every print); and order-book depth. Alpaca's Market Data API serves bars, quotes, trades and news.
- 1.6 Developing Hypotheses for Quant and Algorithmic Trading Models (A1, K1): Every trading model starts with a hypothesis: a precise, testable statement about market behaviour - not 'buy good stocks' but 'when SPY closes below its lower Bollinger Band, its average 5-day forward return exceeds its unconditional 5-day return'.

Facilitation cues

- Open with the success stories (Thorp, Renaissance, D.E. Shaw, Man AHL, Two Sigma, the quant market makers) and ask: 'What did they all have in common?' - a testable hypothesis, clean data, out-of-sample testing, costs and risk control.
- Walk the evolution of quant as a chain: probability and option pricing -> factor models -> statistical arbitrage -> HFT speed race -> ML -> LLMs and agents. Tie each failure to a control used later: LTCM -> leverage limits; quant quake -> crowding; Knight Capital -> deployment testing and the kill switch.
- Define agentic AI as the capability (tools, skills, memory, instructions) and an AI agent as its special always-on class that stands by 24/7 (OpenClaw, Hermes Agent, OpenAI Dots).
- On the ML-to-LLM shift, stress the new backtest risk: an LLM may 'know' its training period, so LLM signals must be tested after the knowledge cutoff.

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- Insist that every hypothesis in Lab 4 states all five elements before any data is examined. This is assessed in PP Task 1.

Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account

Item	Facilitation
TSC / duration / slide	K5 | 20 min | slide 67
Lab manual	labs_lowcode/Lab01-alpaca-setup/Lab01_Instructions.pdf
Objective	Get the simulator running, decide between the offline demo and an Alpaca paper account, and check every integration's status.
In the Trading Simulator	Step 1. Load the demo data (settings) Step 2. Check the integrations (integrations) Step 3. Check the AI agent login (integrations) Step 4. Read the market clock (trading) Step 5. Tour the Pipeline Map (pipeline)
Expected in the simulator	A working app with demo data loaded, a verified Alpaca paper account or the simulator, and a known AI-agent brain.
Debrief prompt	A trading pipeline is only as safe as its configuration and its connection to the broker. One validated config file, paper-only enforcement, keys held in the environment and a pre-flight check that a scheduler can read from the exit code show how AI and algorithmic trading programs are connected to execution platforms in a controlled way (K5).

Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView

Item	Facilitation
TSC / duration / slide	A3, K5, K9 | 20 min | slide 69
Lab manual	labs_lowcode/Lab02-tradingview-charts/Lab02_Instructions.pdf
Objective	Prove the app's prices are right: compare its bars with TradingView's Data Window and explain adjusted vs unadjusted prices.
In the Trading Simulator	Step 1. Read TradingView's Data Window (data?tab=reference&ref=AAPL) Step 2. Compare with RAW and ADJUSTED bars (data?tab=reference&ref=AAPL) Step 3. Repeat for SGX:D05 (DBS) (data?tab=reference&ref=D05.SI) Step 4. Inspect the chart (data?tab=chart&symbol=AAPL)
Expected in the simulator	A PASS / FAIL validation for SGX:D05 and NASDAQ:AAPL on 2025-12-31, with the dividend-adjustment factor explained.
Debrief prompt	Charting platforms are analytics tools (A3) and the API is the pipeline's data path (K5); a validation stage proves they agree before anything is automated. Knowing that adjusted and unadjusted prices are different forms of the same data (K9) prevents a classic error: training on adjusted history and then trading on raw quotes without converting.

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Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality

Item	Facilitation
TSC / duration / slide	A6, K9 | 20 min | slide 71
Lab manual	labs_lowcode/Lab03-alpaca-market-data/Lab03_Instructions.pdf
Objective	Ingest bars from Alpaca / Twelve Data / yfinance (or the demo files), run the data-quality checks and score news sentiment.
In the Trading Simulator	Step 1. Fetch data (data?tab=instruments&symbol=AAPL) Step 2. Read the data-quality report (data?tab=quality&symbol=SPY) Step 3. Switch timeframes (data?tab=chart&symbol=SPY&timeframe=1Week) Step 4. Headlines and sentiment (data?tab=news&symbol=NVDA)
Expected in the simulator	A cached universe, a data-quality report and scored headlines with daily sentiment.
Debrief prompt	Ingestion is the first automated stage: it fixes one version of the truth for every later stage and blocks bad data before a model can learn from it. Bars, quotes, trades, news and alternative data are different forms of financial data (K9) with different volume, velocity and variety - which is what turns a trading data set into a big-data application (A6).

Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test

Item	Facilitation
TSC / duration / slide	A1, A6, K1 | 15 min | slide 74
Lab manual	labs_lowcode/Lab04-trading-hypothesis/Lab04_Instructions.pdf
Objective	Write H0 and H1 before looking at results and test them with a Welch t-test and a bootstrap CI.
In the Trading Simulator	Step 1. Write the hypothesis card (research/hypothesis) Step 2. Run the test (research/hypothesis) Step 3. Robustness and multiple testing (research/hypothesis)
Expected in the simulator	A decision (SUPPORTED / NOT SUPPORTED) with p-value, edge in bps and a robustness grid.
Debrief prompt	Writing the hypothesis before testing, with a null, a cost-aware decision rule and a count of the variants tried, is the foundation of data-driven trading (A1). AI and machine-learning methods (K1) only add value when they test well-posed questions, and larger data sets (A6) are how a promising effect in one ETF is confirmed or rejected.

Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca

Day 1 | Slides 77-142 | A2, A3, K3, K5, K7 | Labs 5-9

Learning outcome: Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.

Key teaching points

- 2.1 Technical Indicators: Trend, Momentum, Volatility and Volume (K3): Simple Moving Average (SMA): the average close over n bars. Exponential Moving Average (EMA): a weighted average with smoothing factor 2/(n+1), reacting faster to recent prices. Both are trend indicators.
- 2.2 Classification of Algorithmic Trading and Quant Algorithms (K3, A2): By purpose, trading algorithms fall into seven families: execution, market making, arbitrage, directional (alpha), statistical arbitrage and pairs (hedged), factor and portfolio construction, and hedging and risk control.

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- 2.3 Analytics Platforms and Tools: TradingView versus Alpaca, and Python (K5, A3): TradingView is a charting and strategy-design platform: global markets including SGX, built-in indicators, the Pine Script language, the Strategy Tester for backtests, alerts and webhooks (webhooks need a paid plan), and a built-in Paper Trading account opened from the chart's Trading Panel with your own starting capital and commission.
- 2.4 Coding Languages for Algorithms and Signals, and Order Placement (K7): Pine Script v6 (TradingView's language): every script starts with //@version=6 and declares indicator() or strategy(). Built-in functions such as ta.ema, ta.rsi, ta.crossover and strategy.entry express a signal in a few lines.
- 2.5 Performance and Risk Metric Definitions (K3, A3): Return metrics: total return; compound annual growth rate CAGR = (ending value / starting value)^(1/years) - 1; and the benchmark-relative excess return.

Facilitation cues

- Demonstrate the Pine Editor live: paste the Lab 5 script, add it to the chart, then change one input and show the chart update.
- When explaining metrics, compute one Sharpe ratio by hand on the whiteboard from five daily returns before learners run Lab 9.
- Emphasise order status in Lab 8: a submitted order is not a filled order.
- On the RSI look-back slides, show SPY on 10 October 2025 in the simulator chart with the lower pane on RSI: RSI(2) fell to 5.8 while RSI(14) stayed near 42 - same day, opposite signals, because only the look-back differs.
- Before the strategy families, check that every learner can say what a short position is and why its loss is unlimited; then, on the mean-reversion risk slide, ask which safeguard would have limited the March 2020 losses.
- Learners who finish early: ask them to add an RSI filter to the Lab 6 strategy and compare the Strategy Tester metrics.

Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins

Item	Facilitation
TSC / duration / slide	A3, K3, K7 | 25 min | slide 124
Lab manual	labs_lowcode/Lab05-pine-indicators/Lab05_Instructions.pdf
Objective	Plot SMA/EMA, Bollinger Bands, RSI and MACD and export a Pine Script v6 strategy for TradingView.
In the Trading Simulator	Step 1. Overlay trend indicators (data?tab=chart&symbol=SPY&overlays=sma:50,sma:200,bollinger) Step 2. Oscillators (data?tab=chart&symbol=SPY&osc=macd) Step 3. Export Pine Script v6 (strategies?key=ema_cross)
Expected in the simulator	A chart with overlays and an oscillator pane, and a Pine Script pasted into TradingView whose values match.
Debrief prompt	Technical indicators (K3) turn prices into trend, volatility and momentum readings; defining them once in Pine Script and once in Python, and proving they match (A3), keeps traders and the automated pipeline on the same numbers. Faster settings fire more signals, which means more trades and more cost (K7).

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Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation

Item	Facilitation
TSC / duration / slide	A2, A3, K3, K7 | 30 min | slide 127
Lab manual	labs_lowcode/Lab06-pine-strategy-tester/Lab06_Instructions.pdf
Objective	Run the same MA-crossover in TradingView's Strategy Tester and in the app, reconcile the results, and try TradingView Paper Trading.
In the Trading Simulator	Step 1. Backtest the SMA 50/200 crossover in the app (backtest?strategy=sma_cross&symbols=SPY&fast=50&slow=200&costs=alpaca_retail) Step 2. Run it in TradingView (strategies?key=sma_cross) Step 3. Reconcile and paper trade on TradingView (backtest?strategy=sma_cross&symbols=SPY&fast=50&slow=200&costs=alpaca_retail)
Expected in the simulator	Trade counts and net profit that agree within the expected differences (fills, costs, data).
Debrief prompt	Choosing a rule and its parameters is a modelling decision (A2); the Strategy Tester and TradingView Paper Trading are analytics and execution platforms (A3) that report standard metrics (K3). An independent reconciliation is the gate between a chart prototype and the automated pipeline, and costs typed into strategy() (K7) barely touch a slow rule but compound for a fast one.

Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals

Item	Facilitation
TSC / duration / slide	A2, K3, K7 | 25 min | slide 131
Lab manual	labs_lowcode/Lab07-technical-indicators/Lab07_Instructions.pdf
Objective	Pick and customise one strategy from each family and compare them after costs.
In the Trading Simulator	Step 1. Trend: EMA 20/50 on SPY (backtest?strategy=ema_cross&symbols=SPY&fast=20&slow=50&costs=alpaca_retail) Step 2. Mean reversion: Bollinger on KO (backtest?strategy=bollinger_reversion&symbols=KO&window=20&k=2) Step 3. Momentum and breakout (backtest?strategy=donchian_breakout&symbols=QQQ&entry=20&exit=10) Step 4. Read each strategy's explanation (strategies)
Expected in the simulator	Four backtests with tuned parameters and an explanation of when each family works.
Debrief prompt	A single signal contract - target positions decided at the close, held from the next bar - is what lets one library feed backtests, live execution and diagnostics (A2). Indicators (K3) become signals through simple, testable rules, and the number of entries each rule produces is the first estimate of its trading cost (K7).

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Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session

Item	Facilitation
TSC / duration / slide	A3, K5 | 30 min | slide 134
Lab manual	labs_lowcode/Lab08-paper-orders/Lab08_Instructions.pdf
Objective	Place market, limit, stop and bracket paper orders, including an extended-hours limit order, and follow each order's lifecycle.
In the Trading Simulator	Step 1. Market order in the regular session (trading?tab=ticket) Step 2. Extended-hours limit order (Singapore class hours) (trading?tab=ticket) Step 3. Bracket order (trading?tab=ticket) Step 4. Queue while closed and cancel (trading?tab=ticket)
Expected in the simulator	A blotter with filled, working, queued and cancelled orders and the risk checks for each.
Debrief prompt	Order types are tools for price, timing and risk, and the trading session decides which ones the broker accepts (K5). An order manager that reads the clock, prices limits off the quote, makes every order idempotent and prints its lifecycle is how an automated program knows exactly what happened at the broker (A3) - in New York's regular session or in its overnight session during a Singapore workday.

Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

Item	Facilitation
TSC / duration / slide	A3, K3 | 25 min | slide 138
Lab manual	labs_lowcode/Lab09-performance-metrics/Lab09_Instructions.pdf
Objective	Read the full metrics table and compare a strategy with buy-and-hold.
In the Trading Simulator	Step 1. Open the analytics (analytics) Step 2. Compare with buy-and-hold (analytics) Step 3. Glossary (help?tab=glossary)
Expected in the simulator	A metrics table you can explain line by line.
Debrief prompt	Performance metrics (K3) answer different questions: CAGR measures growth, Sharpe and Sortino reward per unit of risk, drawdown and Calmar the pain an investor must survive, beta how much is just the market. Running every strategy through one evaluation stage (A3) is what makes the comparison fair.

Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals

Day 2 | Slides 143-171 | A4, A5, K2, K4, K6 | Labs 10-14

Learning outcome: Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.

Key teaching points

- 3.1 Types of Statistical Analyses, Data Models, Algorithms and Computational Methods (K2, A2): Descriptive analysis summarises what happened (mean return, volatility, skew, correlation); inferential analysis tests whether an effect is real (t-test, bootstrap); predictive analysis forecasts (regression, classification); prescriptive analysis decides (optimisation).
- 3.2 Supervised Learning Models for Trading Signals (K4, A4): Target and features: the target is what you predict (next-day direction); the features are what you know at prediction time (lagged returns, RSI, MACD, volatility, volume change). Any feature using future data is leakage.

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- 3.3 Unsupervised Learning: Clustering and PCA for Market Regimes (K4, A4): K-means clustering partitions observations into k groups by minimising within-cluster variance. Applied to rolling return and volatility features it separates calm-bull, choppy and stressed market regimes.
- 3.4 Statistical Modelling Techniques for Financial Time Series (K6, A4): Returns, not prices: model log or simple returns, which are closer to stationary. Prices trend and wander; regressions on raw prices produce spurious results.
- 3.5 A/B Testing Trading Strategies (A5, K2): An A/B test compares variant A (for example a 20/50 crossover) with variant B (a 10/30 crossover, or the same rule with an RSI filter) on the same data and period, to decide whether B is genuinely better.
- 3.6 Pairs Trading: Statistical Arbitrage as a Hedging Algorithm (K6, A4): Pairs trading holds one stock long and an economically linked stock short, in hedge-ratio proportion. Market moves that hit both legs largely cancel, so the position is a hedge: its profit or loss comes from the spread between the two, not from the direction of the market.

Facilitation cues

- Show the strategy-search heatmap from Lab 10 and ask: 'Which cell would you trade?' Then show out-of-sample decay - the best in-sample cell is rarely best afterwards.
- In Lab 11, accuracy near 50-55% is normal for daily direction; the point is the net-of-cost strategy, not the accuracy.
- For the A/B test, make learners write H0, H1 and alpha BEFORE running Lab 13. This is a CRITICAL marking point in PP Task 3.
- In Lab 14 show that JPM/BAC is 93% correlated yet NOT cointegrated, and that the KO/PEP hedge leaves almost no market beta even though the pair lost money after 2022 - the hedge works, the edge did not persist.

Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

Item	Facilitation
TSC / duration / slide	A2, A4, K2 | 20 min | slide 161
Lab manual	labs_lowcode/Lab10-strategy-grid-search/Lab10_Instructions.pdf
Objective	Grid-search SMA crossover parameters in-sample and measure how the ranking decays out of sample.
In the Trading Simulator	Step 1. Run the grid (research/grid?strategy=sma_cross&symbol=SPY) Step 2. Read the decay (research/grid) Step 3. Walk-forward (research/walkforward?strategy=sma_cross)
Expected in the simulator	IS and OOS Sharpe heatmaps, a rank correlation and a robust parameter region.
Debrief prompt	Grid search is a statistical search over model structures (A2, A4); comparing in-sample with out-of-sample results (K2) exposes how much of the 'best' result was luck. A hold-out period, a count of trials and a preference for robust regions are the pipeline's defences against fooling itself.

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Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position

Item	Facilitation
TSC / duration / slide	A4, K4, K6 | 20 min | slide 163
Lab manual	labs_lowcode/Lab11-ml-signal-classifier/Lab11_Instructions.pdf
Objective	Train logistic regression, random forest and XGBoost walk-forward and trade P(up) > threshold.
In the Trading Simulator	Step 1. Train the classifiers (research/ml) Step 2. Backtest an ML strategy (backtest?strategy=ml_xgboost&symbols=SPY) Step 3. Check for leakage (research/diagnostics?strategy=ml_logistic)
Expected in the simulator	Accuracy / AUC table, feature importance and a net-of-cost backtest of the ML signal.
Debrief prompt	Supervised models (K4) can be trained and validated correctly (K6) and still find no edge - an honest negative result is valuable evidence (A4). Walk-forward validation stops the model learning from the future, a leakage test proves the check works, and the cost-aware backtest shows what an AUC of 0.5 is worth in money.

Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

Item	Facilitation
TSC / duration / slide	A4, K4, K6 | 15 min | slide 165
Lab manual	labs_lowcode/Lab12-market-regimes/Lab12_Instructions.pdf
Objective	Find market regimes with walk-forward K-means and gate a signal in the high-volatility regime.
In the Trading Simulator	Step 1. Analyse regimes (research/regime) Step 2. Backtest the regime-filtered strategy (backtest?strategy=regime_filtered&symbols=SPY)
Expected in the simulator	PCA explained variance, a regime timeline and gated vs ungated performance.
Debrief prompt	Unsupervised learning (K4) finds structure without labels: PCA compresses correlated assets into a few drivers and K-means groups days into regimes. Fitting the regimes walk-forward is part of validating the model (K6), and testing the gate signal by signal after costs (A4) shows that a filter can help one strategy and hurt another.

Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves

Item	Facilitation
TSC / duration / slide	A5, K2 | 15 min | slide 167
Lab manual	labs_lowcode/Lab13-ab-testing/Lab13_Instructions.pdf
Objective	Decide statistically between two variants before moving capital.
In the Trading Simulator	Step 1. Run the A/B test (research/abtest?a=trend_filter&b=dip_buy) Step 2. Your own variants (research/abtest?a=ema_cross&b=ema_cross)
Expected in the simulator	Welch and paired t-tests, a bootstrap Sharpe CI and a decision with a multiple-testing caution.
Debrief prompt	A/B testing (A5) applies statistical inference to a business choice: the trend filter looks better, but the evidence cannot reject 'no difference'. Statistical tests (K2) - paired where the data are paired, bootstrapped where returns are not normal, corrected for the number of variants tried - protect capital from being moved on noise.

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Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve

Item	Facilitation
TSC / duration / slide	A4, K6 | 20 min | slide 169
Lab manual	labs_lowcode/Lab14-pairs-trading/Lab14_Instructions.pdf
Objective	Screen pairs for cointegration and backtest a market-neutral KO/PEP spread.
In the Trading Simulator	Step 1. Screen the pairs (research/pairs) Step 2. Backtest KO/PEP (backtest?strategy=pairs_cointegration&symbols=KO,PEP)
Expected in the simulator	An Engle-Granger screen and a pairs backtest with beta near zero.
Debrief prompt	Pairs trading is a hedging algorithm: the short leg cancels most of the market exposure of the long leg, which is why the beta to SPY is near zero (A4). The statistical model is cointegration, not correlation (K6). A pair that is cointegrated in one period can stop reverting in the next, so professional desks re-test cointegration on a rolling basis and size each pair small.

Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis

Day 2 | Slides 172-196 | A7, A8, A9, A10, K7, K8 | Labs 15-20

Learning outcome: Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.

Key teaching points

- 4.1 Architecture of an End-to-End Algorithmic Trading Pipeline (A7, K7): The pipeline: (1) strategy search and research, (2) data ingestion, (3) signal generation, (4) backtest with costs, (5) risk checks, (6) order execution, (7) forward test on paper, (8) monitoring, logging and performance reporting.
- 4.2 Backtesting with Realistic Trading Costs (A8, K7): A backtest replays a strategy on historical data. Vectorised backtests are fast (whole-array operations); event-driven backtests replay bar by bar and model orders more realistically.
- 4.3 Forward Testing with Automated Paper Trading (A7, A8): Forward testing (paper trading) runs the strategy in real time on data it has never seen, placing simulated orders - the bridge between a backtest and live capital.
- 4.4 Potential Reasons for Unintended Outcomes and How to Diagnose Them (A9, K8): Overfitting: parameters tuned to noise. Signs - a spectacular in-sample Sharpe that collapses out of sample, and performance that is very sensitive to small parameter changes.
- 4.5 Performance Analysis, Insights and Reporting (A10): A tearsheet presents the equity curve against the benchmark, the drawdown chart, monthly returns, rolling Sharpe ratio, the metrics table and the cost breakdown on one page.

Facilitation cues

- Draw the pipeline on the whiteboard first; every Topic 4 lab is one box of it.
- In Lab 16, run the bot once together in offline replay, then let learners with keys run it against their paper account during market hours if the class time overlaps US trading hours (21:30-04:00 SGT, 22:30-05:00 in northern winter).
- Lab 18 contains seeded faults: do not reveal them. Prompt with 'Is this result too good?' and 'On which bar was this order actually filled?'

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- In Lab 20 stress that the LLM never bypasses the coded risk manager, and that every decision is logged - the governance message of Topic 1, implemented.

Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even

Item	Facilitation
TSC / duration / slide	A7, A8, K7 | 25 min | slide 184
Lab manual	labs_lowcode/Lab15-trading-costs/Lab15_Instructions.pdf
Objective	Run the same strategy under four cost models and find its break-even cost.
In the Trading Simulator	Step 1. Zero vs Alpaca vs high-cost (backtest? strategy=dip_buy&symbols=SPY&costs=high_cost) Step 2. Break-even and execution timing (backtest? strategy=dip_buy&symbols=SPY&execution=next_close) Step 3. Stops and sizing (backtest? strategy=ema_cross&symbols=SPY&fast=20&slow=50&stop_loss_pct=5&sizing=vol_target)
Expected in the simulator	Gross vs net CAGR, a cost breakdown and the break-even bps.
Debrief prompt	Commission is only one trading cost (K7): spread and slippage scale with turnover, and the assumed fill time can matter as much as the fee schedule. Coding the cost model into the engine (A7) and testing outcomes under several scenarios (A8) shows whether an edge survives real-world execution - the break-even cost is the single number that says how much room the strategy has.

Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading

Item	Facilitation
TSC / duration / slide	A7, A8, K7 | 25 min | slide 186
Lab manual	labs_lowcode/Lab16-paper-trading-bot/Lab16_Instructions.pdf
Objective	Create a bot, run it on a schedule, watch orders, positions and P&L live, and halt it with the kill switch.
In the Trading Simulator	Step 1. Create the bot (trading?new=1&strategy=ema_cross) Step 2. Start it and watch (trading) Step 3. Kill switch (trading) Step 4. Backtest vs forward (trading)
Expected in the simulator	A bot equity curve, a blotter of its orders, a backtest-vs-forward comparison and a tested kill switch.
Debrief prompt	The runner is a software agent coded from a selected model (A7): signal on completed bars, deterministic risk limits, idempotent execution, a kill switch and an audit log. The forward test and the live paper loop check the actions it takes and their outcomes on unseen data (A8), including the real effect of whole shares, fills and session rules (K7).

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Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution

Item	Facilitation
TSC / duration / slide	A7, K5, K7 | 20 min | slide 188
Lab manual	labs_lowcode/Lab17-tradingview-webhook/Lab17_Instructions.pdf
Objective	Send TradingView alerts to the app's webhook and see them pass validation, signal confirmation and risk into paper orders.
In the Trading Simulator	Step 1. Copy the alert template (integrations) Step 2. Run the self-test (integrations) Step 3. Real alerts (paid TradingView plan) (integrations)
Expected in the simulator	7 simulated alerts handled as expected and the audit log.
Debrief prompt	A webhook lets a charting platform trigger execution on a trading platform (K5), but the gateway code (A7) is where the controls live: authentication, allow-lists, size limits, duplicate protection, the pipeline's own signal, the risk layer and an audit log. Each automated order still pays spread and slippage (K7), so alert frequency matters.

Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live

Item	Facilitation
TSC / duration / slide	A9, K8 | 20 min | slide 190
Lab manual	labs_lowcode/Lab18-diagnose-bias/Lab18_Instructions.pdf
Objective	Run the automated diagnostics on a deliberately biased strategy, fix it, and re-run.
In the Trading Simulator	Step 1. Diagnose the biased exhibit (research/diagnostics?strategy=custom_lab18_biased_smoothed_momentum) Step 2. Fix it in the Strategy Builder (builder?key=custom_lab18_biased_smoothed_momentum) Step 3. Re-run diagnostics (research/diagnostics?strategy=custom_lab18_biased_smoothed_momentum)
Expected in the simulator	A FAIL on the biased exhibit, a PASS / WARN table for the fixed version and a verdict.
Debrief prompt	Most 'too good to be true' results come from a short list of causes (K8): look-ahead in features or execution, overfitting, survivorship and ignored costs. Automating the checks and using the verdict as a gate (A9) is what stops an unintended outcome from reaching even a paper account - and the same checks run on the desk's own strategy, not only on other people's.

Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report

Item	Facilitation
TSC / duration / slide	A8, A10 | 25 min | slide 192
Lab manual	labs_lowcode/Lab19-performance-report/Lab19_Instructions.pdf
Objective	Produce a one-page tearsheet and explain the gap between backtest and forward test.
In the Trading Simulator	Step 1. Export the tearsheet (analytics) Step 2. Backtest vs forward (analytics) Step 3. Draw insights (analytics)
Expected in the simulator	A PDF tearsheet and a written explanation of the backtest-vs-forward differences.
Debrief prompt	Comparing the backtest with the forward test is the final test of whether actions and outcomes match the model (A8). Automated monitoring with explicit thresholds turns that comparison into an alarm, and a good report turns numbers into trends, insights and a clear, conditional recommendation (A10) - including the honest message that the strategy reduces risk but has not beaten buy and hold.

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Lab 20: Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval

Item	Facilitation
TSC / duration / slide	A7, A10, K1 | 20 min | slide 194
Lab manual	labs_lowcode/Lab20-agentic-capstone/Lab20_Instructions.pdf
Objective	Run the whole pipeline end to end, get a human-approved proposal from the multi-agent desk, switch on the always-on agent, and create, backtest and forward-test your own strategy (from a template or written for you by AI).
In the Trading Simulator	Step 1. Run the multi-agent desk (agents) Step 2. Human approval (agents) Step 3. Always-on agent (agents) Step 4. Create your own strategy (builder?template=ema_rsi) Step 5. Backtest, diagnose and forward-test it (trading?new=1) Step 6. End-to-end on the Pipeline Map (pipeline)
Expected in the simulator	A Pipeline Map with every stage green, an approved agent proposal with its decision trace, and your custom strategy trading on paper.
Debrief prompt	Agentic AI (K1) chains specialised agents that reason, debate and act, which makes it powerful and risky. Wiring it into an end-to-end pipeline whose gates, deterministic risk limits, kill switch and human approval it cannot override (A7), and evaluating every stage's evidence critically (A10), is how a desk adopts AI and automation responsibly.

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5. Assessment Administration

The Assessment Plan is the governing document for the assessment; this section summarises what the trainer-assessor does on the day.

Component	Duration	Coverage	Conditions	Ratio
Written Assessment (Short Answer Questions)	45 min	9 questions, K1-K9	Open book, individual, invigilated	1 : 20
Practical Performance	75 min	4 tasks, A1-A10	Open book, individual, own laptop with the Trading Simulator; configure and run the provided program - no coding	1 : 20

Sequence on the day

1. Verify each candidate's identity; candidates sign the assessment attendance record.
2. Brief candidates (5 minutes, not assessed): components, durations, open book, individual work, pass rule, appeal and re-assessment.
3. Issue the Written Assessment question paper (Assessment Plan Annex A); start the 45-minute clock; collect scripts at time.
4. Copy courseware/assessment/pp_assessment/candidate into each candidate's labs_lowcode/pp_assessment/ (never the assessor folder), then issue the Practical Performance question paper (Annex C); start the 75-minute clock. Observe and note process evidence.
5. Collect the candidate's pp_config.yaml, pp_output.txt, screenshot and completed response template (saved as PP_<Name>_<Date>).
6. Mark against the marking points (Annex B / D / E), apply the Decision Rules and the Assessor Key (Annex F), derive bands, complete the Score Summary and the Individual Assessment Records.
7. Give feedback. A candidate who is Not Yet Competent is counselled on the gaps and offered re-assessment on the component(s) not met.

Pass rule

- (1) Written Assessment: at least 80 of 100 marks.
- (2) Practical Performance: at least 80 of 100 marks.
- (3) Every knowledge statement (K1-K9) and every ability statement (A1-A10) scores at least 60% of the marks allocated to it (table below), so that each statement is demonstrated.
- (4) Every CRITICAL marking point (1.1, 3.5, 4.1, 4.5, 4.9) is awarded.

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6. Assessor's Keys

The assessor's keys, expected values and the marking rubric are trainer-only and are held in the full Facilitator Guide and the Assessment Plan.

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